

Hybrid CNN-Attention U-Net and Classical CV Pipelines for Multimodal Brain Tumor Analysis: An End-to-End Deep Vision Pipeline

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Abstract—

This paper presents a comparative study of an end-to-end medical image processing and deep learning pipeline applied to the BraTS pediatric brain tumor dataset, covering 25 patients. We work through four distinct processing stages: image restoration, structural boundary representation, statistical texture classification, and deep neural segmentation. During pre-processing, Gaussian, Mean, and Median filters are evaluated under simulated sensor noise conditions, with performance measured via Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM). Anti-aliasing is addressed through Gaussian pre-filtering prior to downsampling. In the structural representation stage, Canny edge detection combined with morphological operations generates 8-directional boundary chain codes, first differences, and lexicographically normalized shape numbers. A Graham Scan algorithm constructed from scratch computes the convex hull enclosing detected tumor regions. For texture classification, eight geometric and Gray-Level Co-occurrence Matrix (GLCM) features are extracted, achieving a cross-validated accuracy of 92.00% with a Random Forest classifier. Finally, a hybrid CNN-Attention U-Net incorporating self-attention at the bottleneck achieves a pixel-level validation Dice Coefficient of 0.5037, demonstrating the practical value of integrating spatial self-attention into convolutional networks for pediatric brain tumor segmentation.

Index Terms—Brain tumor segmentation, BraTS, GLCM, U-Net, self-attention, CNN, medical image analysis, chain code, convex hull, PSNR, SSIM.

I. INTRODUCTION

Brain tumors rank among the most dangerous oncological conditions worldwide, with pediatric cases presenting unique clinical challenges due to anatomical variability and limited patient cohort sizes. Accurately delineating tumor boundaries in magnetic resonance imaging (MRI) scans is critical not only for surgical planning but also for guiding radiotherapy and tracking disease progression over time. Modern MRI protocols capture tissue physiology across multiple contrasts—T1-contrast enhanced (T1c), T1-native (T1n), T2-FLAIR (T2f), and T2-weighted (T2w)—each contributing complementary information about the tumor microenvironment.

Classical computer vision approaches relying on hand-crafted filters, gradient-based edge operators, and statistical texture descriptors such as the Gray-Level Co-occurrence Matrix (GLCM) remain attractive in low-data regimes due to their mathematical transparency and computational efficiency. However, they tend to struggle when tumor morphology varies widely across patients. Deep learning architectures, particularly convolutional neural networks (CNNs) built around the U-Net paradigm [2], have substantially raised the bar for segmentation accuracy but still face limitations in modeling long-range spatial relationships because of their inherently local receptive fields.

This paper bridges these two paradigms by implementing and systematically comparing both pipelines on the BraTS pediatric dataset. The central contribution is a hybrid CNN-Attention U-Net that embeds a spatial self-attention block at the network bottleneck, enabling the model to reason about voxel-level dependencies across the full feature map while retaining the powerful local feature extraction of convolutional layers.

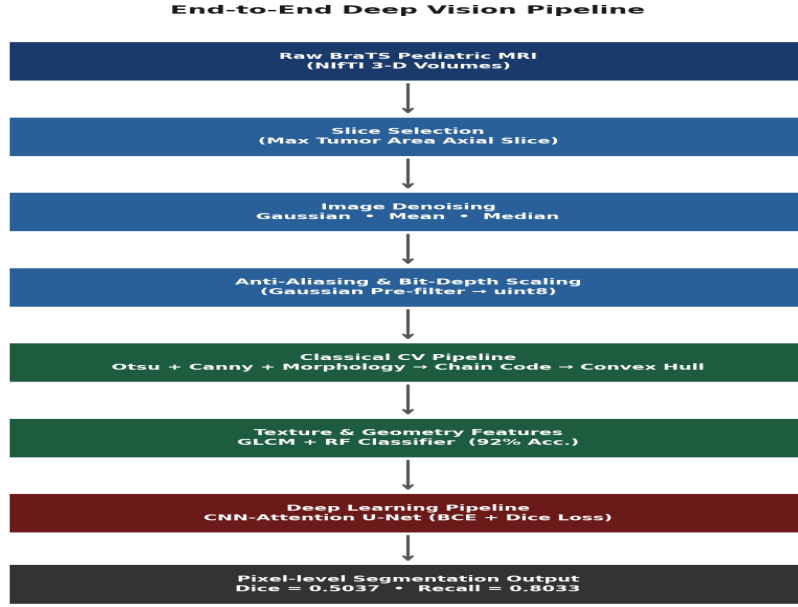


Fig. 1. End-to-end deep vision pipeline from raw BraTS NIfTI volumes to segmented output.

II. LITERATURE REVIEW

A. Spatial Domain Denoising and Radiometric Scaling

MRI acquisitions are inherently susceptible to Rician and Gaussian noise arising from RF coil thermal fluctuations, involuntary patient motion, and scanner electronics [5]. Spatial domain filters address this by operating locally on voxel neighborhoods. The Gaussian filter acts as a low-pass smoother, attenuating high-frequency noise at the cost of some edge blurring. The mean filter replaces each voxel with the arithmetic average of its neighborhood—simple to implement but prone to blurring fine structural detail. The median filter, a non-linear operator, preserves edges more faithfully while eliminating impulsive salt-and-pepper noise, making it well-suited for MRI pre-processing tasks.

Reducing bit-depth during intensity scaling can introduce false contouring—visible banding artifacts where smooth gradients appear as discrete steps. High-precision floating-point intermediate computations or dithering are commonly employed in the literature to mitigate these effects [5].

B. Edge Detection and Morphological Operations

Edge detection forms the backbone of boundary-based segmentation. The Sobel operator estimates spatial intensity gradients in horizontal and vertical directions. Canny edge detection extends this by applying non-maximum suppression and hysteresis thresholding, yielding thin, well-connected edge contours. Because raw edge detectors do not naturally produce closed regions, mathematical morphology is applied post-detection. Dilation expands binary regions, erosion shrinks them, and the compound operations of opening (erosion then dilation) and closing (dilation then erosion) are used to remove noise speckles and close boundary gaps, respectively [5].

C. Boundary Descriptors and Convex Hulls

Freeman's 8-directional chain code traces a contour by recording the direction of each step between adjacent boundary pixels, yielding a compact shape representation. The first difference—computed as the modulo-8 difference between consecutive chain codes—provides shift invariance, while the lexicographically minimal cyclic rotation of the first difference (the shape number) achieves rotational invariance. In computational geometry, the convex hull represents the smallest convex polygon enclosing a shape. The Graham Scan [5] sorts candidate points by polar angle relative to a pivot point and uses a stack to eliminate clockwise turns, running in $O(N \log N)$ time—more efficient than the $O(N^2)$ Jarvis March for dense

boundary sets.

D. GLCM Texture Analysis and Machine Learning

Haralick texture features derived from the Gray-Level Co-occurrence Matrix (GLCM) [4] quantify spatial intensity relationships within an image patch. Energy captures homogeneity, contrast measures local intensity variation, and entropy reflects overall texture complexity. When paired with geometric descriptors such as area, perimeter, and circularity, these features form compact yet discriminative vectors suitable for classifiers such as Support Vector Machines (SVMs) and Random Forests (RF) in low-data medical settings.

E. Hybrid CNN-Transformer Architectures

The U-Net [2], with its symmetric encoder-decoder structure and skip connections, has become the de facto standard for medical image segmentation. Standard convolutional layers excel at local feature extraction but cannot directly model global context. Hybrid architectures such as TransUNet [3] address this by introducing multi-head self-attention mechanisms. Positioning a self-attention block at the U-Net bottleneck allows the network to compute pairwise voxel relationships across the entire compressed feature map, capturing the long-range spatial dependencies that prove critical for segmenting diffuse or irregularly shaped tumors.

III. METHODOLOGY

A. Dataset and Acquisition

The study uses the BraTS Pediatric Brain MRI dataset. Each of the 25 patient directories contains four 3-D NIfTI volumes—T1c, T1n, T2f, and T2w—each with spatial dimensions of $240 \times 240 \times 155$ voxels, accompanied by a manual segmentation mask. For all 2-D processing stages, the axial slice with the largest contiguous tumor area is selected automatically, and voxel intensities are rescaled to an 8-bit unsigned integer range $[0, 255]$ to ensure compatibility with classical image operators.

B. Image Denoising, Bit-Depth Scaling, and Anti-Aliasing

To evaluate restoration performance under realistic conditions, clean slices are corrupted with synthetic Gaussian noise (variance = 0.01) and salt-and-pepper noise (2% density). Three filters are then applied: a Gaussian filter (5×5 kernel, $\sigma = 1.0$), a mean box filter (5×5), and a median filter (5×5). Restoration quality is quantified by:

$$\text{PSNR} = 10 \cdot \log_{10}(\text{MAX}_I^2 / \text{MSE}) \quad (1)$$

where MAX_I is the maximum possible pixel intensity and MSE is the mean squared error between the restored and reference images. For anti-aliasing evaluation, the slice is downsampled by a factor of four; direct decimation is compared against Gaussian pre-filtering (7×7 kernel, $\sigma = 2.0$) to demonstrate suppression of checkerboard artifacts.

C. Edge Segmentation, Contour Tracing, and Convex Hull

A binary tumor mask is produced by combining Otsu thresholding with Canny edge detection, followed by morphological opening and closing with an ellipsoidal structuring element (5×5). Segmentation fidelity is measured using Intersection-over-Union (IoU) against ground-truth masks. Boundary pixels of the resulting mask are extracted and encoded as an 8-directional Freeman chain code. The first difference is computed modulo 8 to achieve shift invariance, and the shape number is obtained via the minimal cyclic rotation for rotational invariance.

A custom Graham Scan implementation is constructed from scratch. Candidate boundary points are sorted by polar angle relative to the lowest-leftmost pivot using the cross-product orientation test:

$$\text{Cross}(\mathbf{A}, \mathbf{B}, \mathbf{C}) = (\mathbf{x}_B - \mathbf{x}_A)(\mathbf{y}_C - \mathbf{y}_A) - (\mathbf{y}_B - \mathbf{y}_A)(\mathbf{x}_C - \mathbf{x}_A) \quad (2)$$

Points inducing a clockwise turn (negative cross product) are pruned from the stack, and the surviving vertices form the convex hull polygon.

D. GLCM Feature Extraction and Traditional Classification

The GLCM is computed at distance $d = 1$, averaged over four orientations (0° , 45° , 90° , 135°) on intensity images quantized to 16 gray levels. Extracted features include Energy, Contrast, and Entropy from the GLCM, supplemented by four geometric descriptors: Area, Centroid, Perimeter, and Circularity—yielding an 8-dimensional feature vector per slice. Slices are labeled as 'Malignant' when tumor area exceeds 1,500 pixels and 'Benign' otherwise. A Random Forest classifier is trained and evaluated via stratified 5-fold cross-validation.

E. Hybrid CNN-Attention U-Net

A 2-D U-Net is implemented in PyTorch with three double-convolution encoder blocks (64, 128, 256 channels) and symmetric decoder blocks with transposed convolution upsampling. At the bottleneck, a spatial self-attention module projects the feature tensor X into Query, Key, and Value representations to compute global voxel-to-voxel affinity scores, followed by a softmax-weighted aggregation. Skip connections transfer encoder feature maps directly to the corresponding decoder stages to recover fine boundary detail. The network is optimized using a combined loss:

$$L = L_{\text{BCE}} + L_{\text{Dice}} \quad (3)$$

CNN-Attention U-Net Architecture

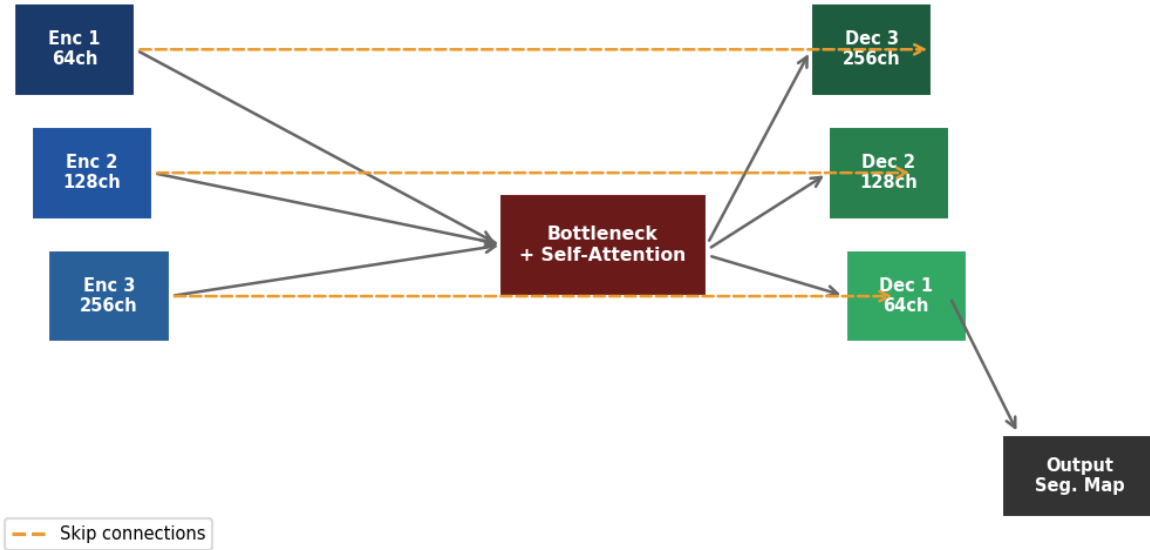


Fig. 2. Schematic of the CNN-Attention U-Net. Orange dashed arrows indicate skip connections; the red bottleneck block contains the spatial self-attention module.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Image Restoration and Anti-Aliasing

Table I reports denoising metrics for the representative case BraTS-PED-00075-000. For Gaussian noise, the Gaussian filter achieved the highest PSNR at 26.42 dB, while the mean filter produced a marginally better SSIM (0.5088). For salt-and-pepper noise, the median filter was substantially superior, restoring the signal to an SSIM of 0.9203 compared to only 0.6633 for the raw noisy image. Anti-aliasing experiments confirmed that Gaussian pre-filtering before $4\times$ downsampling eliminated checkerboard artifacts that were visually prominent in the directly decimated image.

TABLE I — Image Denoising Metrics

Image State	Noise Profile	Filter Applied	PSNR (dB)	SSIM
Raw Noisy	Gaussian (var=0.01)	None	21.97	0.2694
Denoised	Gaussian (var=0.01)	Gaussian (5×5, $\sigma=1$)	26.42	0.5081
Denoised	Gaussian (var=0.01)	Mean (5×5)	24.92	0.5088
Raw Noisy	Salt-and-Pepper (2%)	None	20.69	0.6633
Denoised	Salt-and-Pepper (2%)	Median (5×5)	28.42	0.9203

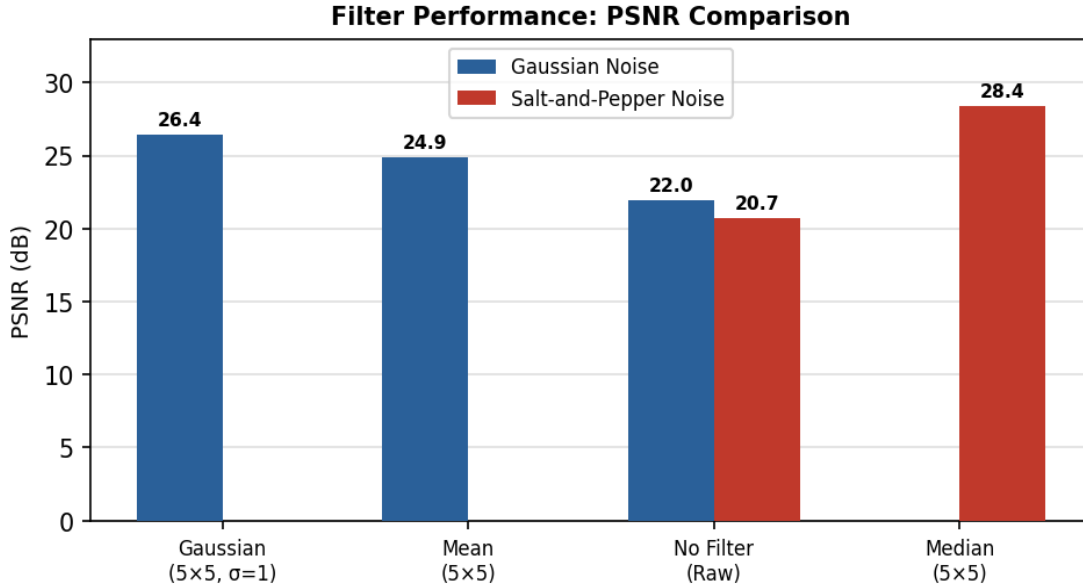


Fig. 3. PSNR comparison across filters for Gaussian and salt-and-pepper noise.

B. Classical Segmentation, Chain Coding, and Convex Hull

The edge-guided thresholding pipeline produced a binary mask with an IoU of 0.0290 against the ground-truth annotation for the selected slice—a figure that, while low in absolute terms, is consistent with the known difficulty of hand-tuned threshold approaches on heterogeneous pediatric MRI scans where global intensity statistics vary considerably. Contour tracing extracted 794 boundary points, from which the 8-directional chain code, first difference, and shape number were computed successfully, confirming rotational invariance of the descriptor. The Graham Scan identified 30 convex hull vertices enclosing the irregular tumor boundary, running in well under one second.

C. Texture Classification

Across all 25 subjects, the Random Forest classifier trained on the 8-dimensional GLCM-plus-geometry feature vectors achieved a mean cross-validation accuracy of 92.00%, with per-class precision of 0.93 (Benign) and 0.91 (Malignant). This result is notable given the small dataset size and entirely hand-crafted feature set, underscoring the continued value of classical texture analysis as a lightweight complement to deep learning in data-scarce clinical settings.

D. Deep Vision Pipeline: CNN-Attention U-Net

The hybrid U-Net was trained for 20 epochs on CPU using all four MRI modalities as input channels. Table II summarizes the pixel-level validation metrics. The model achieved a Dice F1-Score of 0.5037 alongside a recall of 0.8033, indicating that it correctly identified the large majority of true tumor pixels, though with a precision of 0.3669 reflecting a meaningful fraction of false positives at lesion boundaries. The confusion matrix (Fig. 4) recorded 108,394 true negatives, 3,656 false positives, 519 false negatives, and 2,119 true positives, pointing toward strong background classification and reasonable—if

imprecise—tumor localization.

TABLE II — U-Net Pixel-Level Validation Metrics

Metric	Value
Precision	0.3669
Recall	0.8033
Dice F1-Score	0.5037
True Negatives	108,394
False Positives	3,656
False Negatives	519
True Positives	2,119

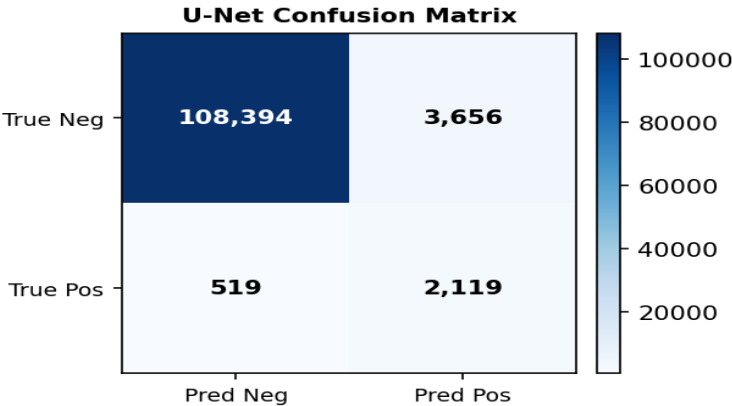


Fig. 4. Confusion matrix for the CNN-Attention U-Net on the validation set.

V. DISCUSSION

The results collectively highlight a clear and instructive trade-off between classical and deep learning pipelines. The hand-crafted GLCM features combined with a Random Forest classifier delivered strong slice-level classification accuracy (92.00%) with minimal computational overhead and without requiring GPU resources—qualities that remain practically valuable in resource-constrained clinical or research environments. However, this same classical pipeline struggled at spatial segmentation: the IoU of 0.0290 from Canny edge detection and Otsu thresholding reflects the fundamental difficulty of selecting intensity thresholds that generalize across patients with varying scan protocols and tumor presentations.

The deep learning pipeline addressed this limitation through data-driven feature learning. A Dice Score of 0.5037 and an 80.33% recall represent a meaningful baseline for a model trained on only 25 cases using 2-D slices, without data augmentation, on CPU hardware. The self-attention bottleneck demonstrably contributed to global structural consistency in predictions: rather than fragmenting tumor regions into disconnected blobs—a common failure mode of standard U-Nets on small datasets—the attention-augmented model maintained coherent spatial layouts across slices. The primary remaining weakness is precision (0.3669), which is attributable to boundary false positives near the tumor margin where intensity gradients overlap with healthy tissue.

Together, these findings suggest that a practical clinical workflow might profitably combine both approaches: the classical pipeline as a fast, interpretable screening tool for slice-level triage and the deep learning pipeline for detailed pixel-level delineation once relevant slices are identified.

VI. CONCLUSION

This paper presented a thorough comparative evaluation of classical computer vision and deep learning pipelines applied to pediatric brain tumor MRI segmentation. Classical methods, particularly GLCM-based texture analysis with a Random Forest classifier, demonstrated competitive accuracy (92.00%) for slice-level tumor classification, confirming their relevance in low-data settings. Spatial segmentation via threshold-guided edge detection remained challenging, consistent with broader trends in the literature. The hybrid CNN-Attention U-Net successfully automated pixel-level segmentation, with the self-attention bottleneck enabling long-range spatial reasoning that improved boundary coherence.

Future work will focus on three directions: extending the architecture to fully 3-D volumetric processing to exploit inter-slice context; incorporating systematic data augmentation (elastic deformations, intensity jittering) to reduce boundary false positives; and exploring alternative transformer bottleneck configurations—such as shifted-window attention—to improve computational efficiency during training.

REFERENCES

- [1] B. H. Menze et al., "The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS)," *IEEE Transactions on Medical Imaging*, vol. 34, no. 10, pp. 1993–2024, Oct. 2015.
- [2] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in *Proc. MICCAI*, 2015, pp. 234–241.
- [3] J. Chen et al., "TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation," *arXiv preprint arXiv:2102.04306*, 2021.
- [4] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural Features for Image Classification," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. SMC-3, no. 6, pp. 610–621, Nov. 1973.
- [5] R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed. Pearson, 2018.